Potential Outcomes and Randomised Experiments

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Preliminaries

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- Accept GitHub invitation (click link in email)
- Upload `.txt` or `.R` or `.qmd` or `.Rmd` of solutions (`.pdf`, but comments)
- How `.qmd`/`.Rmd` works (more to come?)

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- Work together, but every keystroke your own
- Ask questions in the Slack

Review Questions

1. When can we observe both $Y_i(1)$ and $Y_i(0)$?
2. What is this fact called?
3. When does the empirical difference-in-means estimator exactly equal the true, underlying ATE?
4. What are the two parts of SUTVA?

Exercise in Potential Outcomes

Statistical Refreshment

Expectation

$$E(Y) = \sum_{i=1}^{n} [y_i \cdot p(y_i)]$$

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Calculate $E(\text{Ideology})$.

Respondent	Ideology
1	3
2	-2
3	3
4	1
5	1

Expectation

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Calculate $E(\text{Ideology})$.

Respondent	Ideology
1	3
2	-2
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4	1
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$$1 \cdot \frac{2}{5} + 3 \cdot \frac{2}{5} + (-2) \cdot \frac{1}{5} = \frac{6}{5}$$

Conditional Expectation

$$E(Y|X = x) = \sum_{i=1}^{n} [y_i \cdot p(y_i|X = x)]$$

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$$E(Y|X = x) = \sum_{i=1}^{n} [y_i \cdot p(y_i|X = x)]$$

Calculate $E(\text{Ideology}|\text{Dem} = 0)$.

Respondent	Ideology	Dem
1	3	0
2	-2	1
3	3	0
4	1	0
5	1	1

Conditional Expectation

$$E(Y|X=x) = \sum_{i=1}^{n} [y_i \cdot p(y_i|X=x)]$$

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Calculate $E(\text{Ideology}|\text{Dem}=0)$.

Respondent	Ideology	Dem
1	3	0
2	-2	1
3	3	0
4	1	0
5	1	1

$$1 \cdot \frac{1}{3} + 3 \cdot \frac{2}{3} = \frac{7}{3}$$

Conditional Expectation

$$E(Y|X = x) = \sum_{i=1}^{n} [y_i \cdot p(y_i|X = x)]$$

Calculate $E(\text{Ideology}|\text{Dem} = 0)$

		Dem?	
		0	1
	3	2	0
Ideology	1	1	1
	-2	0	1

(tabled data, cells: counts of units with row/column scores)

Conditional Expectation

$$E(Y|X=x)=\sum_{i=1}^{n}[y_i \cdot p(y_i|X=x)]$$

Calculate $E(\text{Ideology}|\text{Dem}=0)$

		Dem?	
		0	1
	3	0.4	0.0
Ideology	1	0.2	0.2
	-2	0.0	0.2

(tabled data, cells: probability unit has row/column scores)

Conditional Expectation

$$E(Y|X=x) = \sum_{i=1}^{n}[y_i \cdot p(y_i|X=x)]$$

Calculate $E(\text{Ideology}|\text{Dem}=0)$

		Dem?	
		0	1
	3	0.4	0.0
Ideology	1	0.2	0.2
	-2	0.0	0.2

(tabled data, cells: probability unit has row/column scores)

$$3 \cdot \frac{0.4}{0.6} + 1 \cdot \frac{0.2}{0.6} = \frac{7}{3}$$

Unbiasedness

Does estimator return the truth, on average?

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Unbiasedness

Does estimator return the truth, on average?

$$E(\hat{\theta}) = \theta$$

- $E\left[\widehat{\overline{Y_1 - Y_0}}\right] = \overline{Y_1 - Y_0}$
- $E\left[\widehat{\beta_1}\right] = \beta_1$

Unbiasedness

What estimator used to estimate $\overline{Y_1 - Y_0}$?

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The difference-in-means estimator.

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$$\widehat{\overline{Y_1 - Y_0}} = (Y_1 | T_i = 1) - (Y_0 | T_i = 0)$$

Unbiasedness

What estimator used to estimate $\overline{Y_1 - Y_0}$?

The difference-in-means estimator.

$$\widehat{\overline{Y_1 - Y_0}} = (Y_1 | T_i = 1) - (Y_0 | T_i = 0)$$

Is it a "good" estimator?

Unbiasedness

The difference-in-means estimator is *unbiased* for the true average treatment effect.

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(See `02-unbiased.R`)

Probability

The Three Axioms

1. $P(A) \geq 0$
2. $P(\Omega) = 1$
3. If events *mutually exclusive* (or, sets *disjoint*), then

$$P(A \text{ or } B) = P(A) + P(B)$$

Conditional Probability

For events A and B,

$$\begin{aligned} P(A \text{ and } B) &= P(A)P(B|A) \\ &= P(B)P(A|B) \end{aligned}$$

Divide both sides by marginal probability $P(B)$ yields

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

Statistical Independence

A and B are independent iff both

$$P(A|B) = P(A)$$

and

$$P(B|A) = P(B)$$

Conditional Independence

A and B can be independent *conditional on* C.

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E.g., Y_1, Y_0 indep of T, given X.

$$P(A \text{ and } B|C) = P(A|C)P(B|C)$$

Potential Outcomes Model: Estimands

- Individual TE

$$\tau_i = Y_i(1) - Y_i(0)$$

- Average treatment effect

$$ATE = E(Y_1 - Y_0) = E(Y_1) - E(Y_0)$$

- Average treatment effect for the treated

$$ATT = E(Y_1|T = 1) - E(Y_0|T = 1)$$

Potential Outcomes Model: Estimands

The average treatment effect (ATE):

$$
\begin{aligned}
E(Y_1 - Y_0) &= \frac{1}{2n}\sum_{i=1}^{2n}(Y_{i1} - Y_{i0}) \\
&= \frac{1}{2n}\sum_{i=1}^{2n}(Y_{i1}) - \frac{1}{2n}\sum_{i=1}^{2n}(Y_{i0}) \\
&= E(Y_1) - E(Y_0)
\end{aligned}
$$

Potential Outcomes Model: Estimands

- If we know (Y_1, Y_0) indep of T
- Then,

$$\begin{aligned} E\left(Y_1\right) &= E(Y_1|T=1) \\ E\left(Y_0\right) &= E(Y_0|T=0) \end{aligned}$$

- Then, can substitute

$$\begin{aligned} ATE &= E(Y_1) - E(Y_0) \\ &= E(Y_1|T=1) - E(Y_0|T=0) \end{aligned}$$

Potential Outcomes Model: Estimands

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- Then, can substitute

$$\begin{aligned} ATE &= E(Y_1) - E(Y_0) \\ &= E(Y_1 | T = 1) - E(Y_0 | T = 0) \end{aligned}$$

Observed diff in Tr and Co group means gives ATE!

Potential Outcomes Model: Estimands

Holland (1986): "*prima facie effect*":

$$E(Y_t|S = t) - E(Y_c|S = c)$$

Potential Outcomes Model: Estimands

Holland (1986): "*prima facie effect*":

$$E(Y_t|S = t) - E(Y_c|S = c)$$

"It is important to recognize that $E(Y_t)$ and $E(Y_t|S = t)$ are *not* the same thing ..."

Potential Outcomes Model: Estimands, Interpretation

Gerber and Green (2012):

When $Y(1)$ and $Y(0)$ indep of T,

$$ATE \quad = \quad E(Y_i(1)|T_i = 1) - E(Y_i(0)|T_i = 0)$$

Potential Outcomes Model: Estimands, Interpretation

Gerber and Green (2012):

When $Y(1)$ and $Y(0)$ indep of T,

$$
\begin{aligned}
ATE &= E(Y_i(1)|T_i = 1) - E(Y_i(0)|T_i = 0) \\
&= E(Y_i(1)|T_i = 1) - E(Y_i(0)|T_i = 1) + \\
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&= [E(Y_i(1)|T_i=1) - E(Y_i(0)|T_i=1)] + \\
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&\quad [E(Y_i(0)|T_i = 1) - E(Y_i(0)|T_i = 0)] \\
&= E(Y_i(1) - Y_i(0)|T_i = 1) + \\
&\quad E(Y_i(0)|T_i = 1) - E(Y_i(0)|T_i = 0)
\end{aligned}
$$

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ATE &= E(Y_i(1)|T_i = 1) - E(Y_i(0)|T_i = 0) \\
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&= E(Y_i(1) - Y_i(0)|T_i = 1) + \\
&\quad E(Y_i(0)|T_i = 1) - E(Y_i(0)|T_i = 0)
\end{aligned}
$$

$$
\underbrace{E(Y_i(1) - Y_i(0)|T_i = 1)}_{\text{ATT}} + \underbrace{E(Y_i(0)|T_i = 1) - E(Y_i(0)|T_i = 0)}_{\text{Selection Bias}}
$$

Assignment of Treatment

What can be a Treatment?

- Timing clearly defined

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- Covariates: causally prior to treatment

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- Covariates: causally prior to treatment
- "Attributes", "immutable characteristics": difficult to isolate

Sen and Wasow (2016) : "Race as a Bundle of Sticks: Designs that Estimate Effects of Seemingly Immutable Characteristics" (*elements* of attributes varyingly manipulable)

Attributes

"Causal effect of race"?

```r
data(resume, package = "qss")
dim(resume)
```

```
[1] 4870    4
```

Attributes

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data(resume, package = "qss")
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[1] 4870    4
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```
kable(table(resume$race, resume$call))
```

	0	1
black	2278	157
white	2200	235

Attributes

"Causal effect of race"?

```r
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dim(resume)
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[1] 4870    4
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Bertrand and Mullainathan (2004): "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination"

Random Assignment Mechanisms

- Simple/complete randomisation
  (Bernoulli trial, prob π)

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- Cluster randomisations
  (assignment at higher level)

Random Assignment Mechanisms

- Simple/complete randomisation
  (Bernoulli trial, prob π)
- Complete randomisation / random allocation
  (fixed proportion to tr)
- Blocked randomisations
  (fixed proportion to tr, w/in group)
- Cluster randomisations
  (assignment at higher level)
- Waitlists, stepped wedges, etc.

The Potential Outcomes Model: Assignment

Observed outcome: $Y_i = Y_i(1) \cdot T_i + Y_i(0) \cdot (1 - T_i)$

The assignment mechanism *selects* which potential outcome we observe.

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Observed outcome: $Y_i = Y_i(1) \cdot T_i + Y_i(0) \cdot (1 - T_i)$

The assignment mechanism *selects* which potential outcome we observe.

(Gerber and Green (2012) use $Y_i = Y_i(1) \cdot d_i + Y_i(0) \cdot (1 - d_i)$ to highlight that we observe pot outcome from treatment **actually** taken, not hypothetical or assigned treatment.)

The Potential Outcomes Model: Assignment

"Assignment mechanisms" are really missing-data-generating procedures.

Ignorability

Assignment mechanism is *ignorable* if Y_{mis} conditnly indep of T

$$P(T|X, Y_{obs}, Y_{mis}) = P(T|X, Y_{obs})$$

Ignorability

Assignment mechanism is *ignorable* if Y_{mis} conditnly indep of T

$$P(T|X, Y_{obs}, Y_{mis}) = P(T|X, Y_{obs})$$

Nothing in unobserved Y_{mis} informs relationship betw'n Y_{obs}, T.

Unconfoundedness

Some ignorable mechanisms are *unconfounded*, too.

$$P(T|X, Y_{obs}, Y_{mis}) = P(T|X)$$

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Nothing in Y informs T.

Unconfoundedness

Some ignorable mechanisms are *unconfounded*, too.

$$P(T|X, Y_{obs}, Y_{mis}) = P(T|X)$$

Nothing in Y informs T.

These are special (key!) cases of conditional independence.

An Assignment Mechanism

Little & Rubin (2000):

Patient	Y	T
1	6	1
2	12	1
3	9	0
4	11	0

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Clearly, treatment is harmful. $\overline{Y(1)} - \overline{Y(0)} = 9 - 10 = -1$.

An Assignment Mechanism

Little & Rubin (2000):

Patient	Y(0)	Y(1)	τ	T
1		6		1
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Mean	10	9		

Clearly, treatment is harmful.

$$\overline{Y(1)|T=1} - \overline{Y(0)|T=0} = 9 - 10 = -1$$

An Assignment Mechanism: Perfect Doctor

Little & Rubin (2000):

Patient	Y(0)	Y(1)	τ	T
1	(1)	6		1
2	(3)	12		1
3	9	(8)		0
4	11	(10)		0
Mean	10	9		

An Assignment Mechanism: Perfect Doctor

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Patient	Y(0)	Y(1)	τ	T
1	(1)	6	(5)	1
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Clearly, treatment is beneficial:

$$\overline{Y(1)} - \overline{Y(0)} = 9 - 6 = 3$$

An Assignment Mechanism: Perfect Doctor

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Mean	10	9	(3)	

Clearly, treatment is beneficial:

$$\overline{Y(1)} - \overline{Y(0)} = 9 - 6 = 3$$

This assg mechanism is non-ignorable, confounded.

Randomisation (Design-based) Inference

A volunteer?

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The task: select the 2 folders with messages

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- What is our baseline expectation/model for this process?

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The task: select the 2 folders with messages

- What is our baseline expectation/model for this process?
  - "No x-ray vision. No ESP. Effect of messages on choice = 0."

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  - "Some way to detect messages. Message location $\rightarrow$ choice."

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  - "No x-ray vision. No ESP. Effect of messages on choice $= 0$."
- What is an alternative?
  - "Some way to detect messages. Message location $\rightarrow$ choice."

Select!

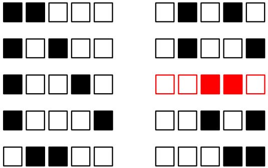
Randomisation Inference

The possible choices:

■■□□□	□■□■□
■□■□□	□■□□■
■□□■□	□□■■□
■□□□■	□□■□■
□■■□□	□□□■■

Randomisation Inference

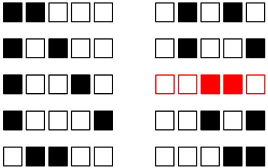
The possible choices:



- You chose _____ and _____. Let X = number found.

Randomisation Inference

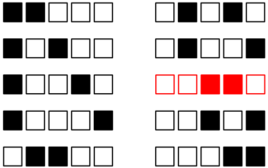
The possible choices:



- You chose _____ and _____. Let X = number found.
- What was $P(X \geq 2|\text{no ESP})$?

Randomisation Inference

The possible choices:



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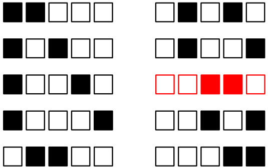

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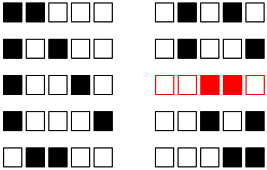
- What was $P(X \geq 1|\text{no ESP})$?

Randomisation Inference

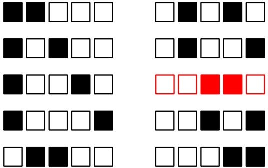
The possible choices:



- You chose _____ and _____. Let X = number found.
- What was $P(X \geq 2|\text{no ESP})$? $\frac{1}{10} = 0.1$
- What was $P(X \geq 1|\text{no ESP})$? $\frac{7}{10} = 0.7$

Randomisation Inference

The possible choices:



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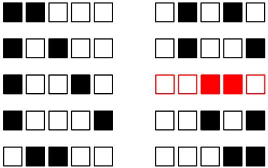
- What is "prob result at least this extreme, given model of no effect"?

Randomisation Inference

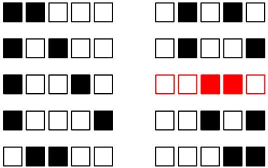
The possible choices:



- You chose _____ and _____. Let X = number found.
- What was $P(X \geq 2|\text{no ESP})$? $\frac{1}{10} = 0.1$
- What was $P(X \geq 1|\text{no ESP})$? $\frac{7}{10} = 0.7$
- What is "prob result at least this extreme, given model of no effect"?
- Definition of p-value!

Randomisation Inference

The possible choices:



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- What was $P(X \geq 2|\text{no ESP})$? $\frac{1}{10} = 0.1$
- What was $P(X \geq 1|\text{no ESP})$? $\frac{7}{10} = 0.7$
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- Definition of p-value!
- Valid, exact, with no distributional assumption, no large n

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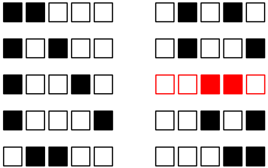

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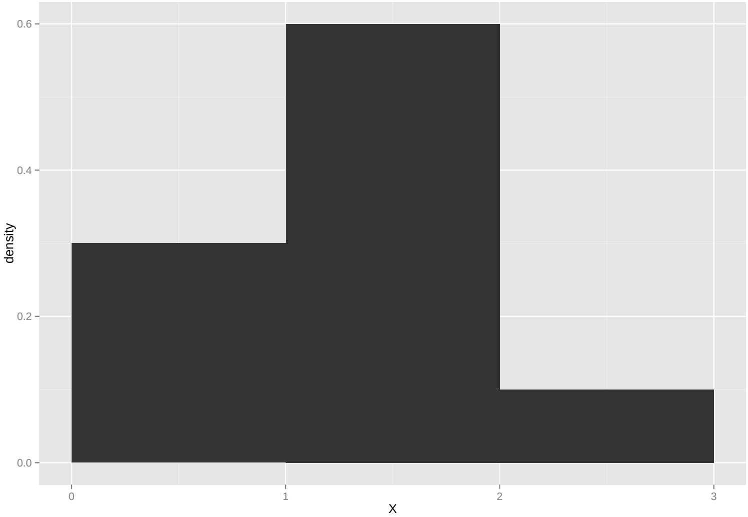
The Randomisation Distribution of X



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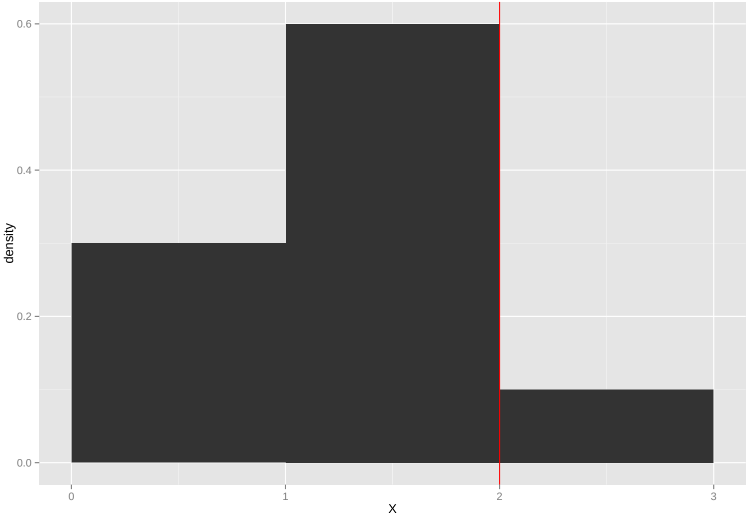

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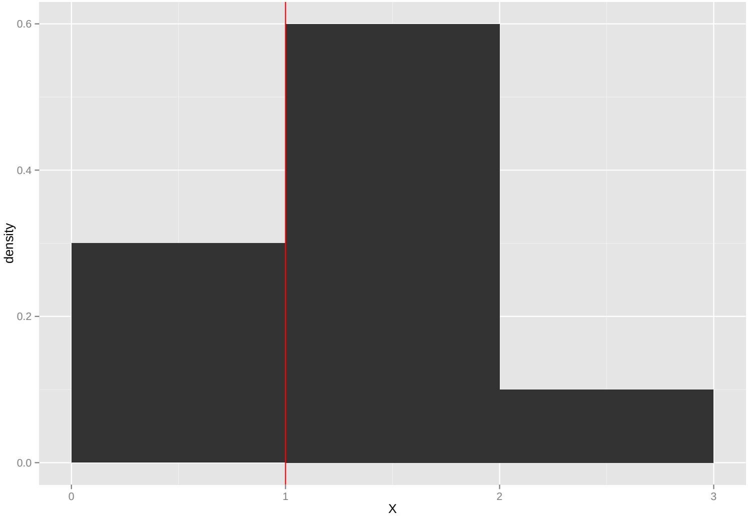

Parametric Null Hypothesis Significance Testing

- Specify and assume H_0
- Define H_A
- Examine reference dist'n (t, χ^2, ...) under H_0
- Calculate p-value
- Compare to some α; reject H_0 if $p < \alpha$

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- CA ballot ordering effects (JASA 2006)

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How many randomisations are there?

Combinations: Counting selected sets

How many ways to **select** k things from a set of n things?

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$$ \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 252 $$

Common Assumptions, Null Hypotheses

- Constant effect:

$$\tau_i = Y_{i1} - Y_{i0} = \tau \quad \forall i$$

- Null hypothesis of no average effect:

$$ATE = \overline{\tau} = 0$$

- Sharp null hypothesis of no effect:

$$\tau_i = 0$$

An Assignment Mechanism: Perfect Doctor

Calculate RI p-value for Perfect Doctor, under sharp null.

Patient	Y(0)	Y(1)	τ	T
1	(1)	6	(5)	1
2	(3)	12	(9)	1
3	9	(8)	(-1)	0
4	11	(10)	(-1)	0
Mean	10	9	(3)	

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(See `02-ri-perfect-dr.R`)

RI versus the t-test

Perfect Doctor:

- RI: $p = 1$
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Perfect Doctor:

- RI: $p = 1$
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(Odd logic of NHST: "assume false thing, how strange is data?")

Randomisation Inference, Example 2

Assumed table of potential outcomes:

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	_	15	10	5
2	_	15	15	0
3	_	30	20	10
4	_	15	20	-5
5	_	20	10	10
6	_	15	15	0
7	_	30	15	15
Average		20	15	5

Randomisation Inference, Example 2

Suppose we randomly select 2 villages to have female-headed councils, and observe

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15		
2	M		15	
3	M		20	
4	M		20	
5	M		10	
6	M		15	
7	F	30		
Average		22.5	16	6.5

Randomisation Inference, Example 2

We assume the *sharp null* hypothesis (assumed values in red):

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15	15	0
2	M	15	15	0
3	M	20	20	0
4	M	20	20	0
5	M	10	10	0
6	M	15	15	0
7	F	30	30	0
Average				0

Randomisation Inference, Example 2

Then we estimate what the observed ATE would be for all the possible random assignments.

First,

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15		
2	M		15	
3	M		20	
4	M		20	
5	M		10	
6	F	15		
7	M		30	
Average		15	19	-4

Randomisation Inference, Example 2

Second,

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15		
2	M		15	
3	M		20	
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5	F	10		
6	M		15	
7	M		30	
Average		12.5	20	-7.5

Randomisation Inference, Example 2

...,

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..., and all the others. The full set of $\frac{7!}{2!5!} = 21$ differences in means:

	Estimate	Frequency
	-7.5	3
	-4	5
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- can be computationally intensive

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```
resume |> group_by(race) |>
  summarise(call_rate = mean(call))
```

```
# A tibble: 2 x 2
  race  call_rate
  <chr>     <dbl>
1 black    0.0645
2 white    0.0965
```

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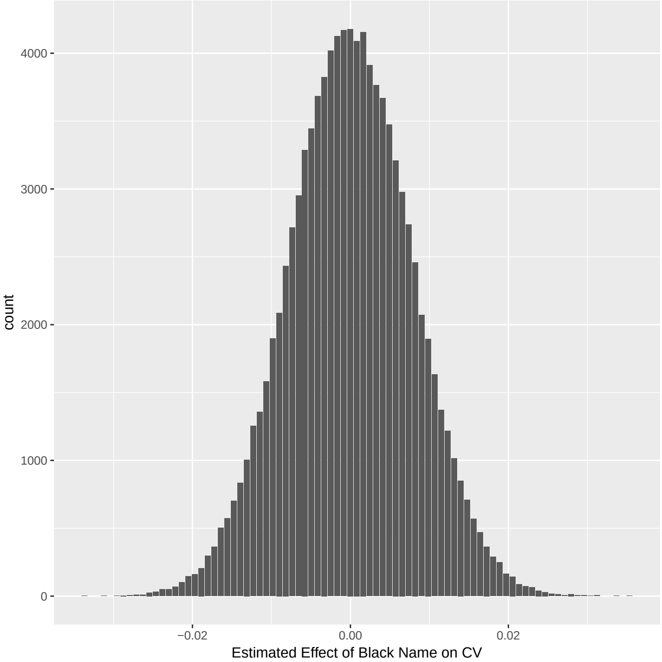
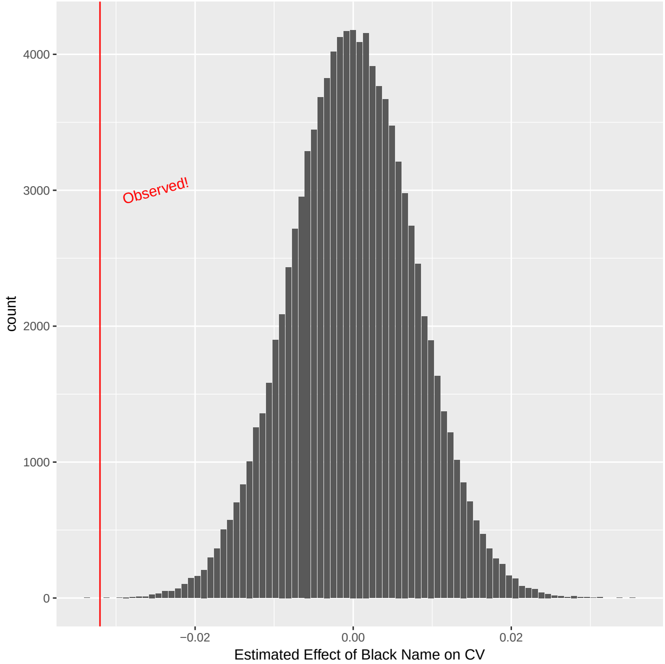
- See `02-ri-resume-donate.R`


4000
3000
2000
1000
0
count
−0.02
0.00
0.02
Estimated Effect of Black Name on CV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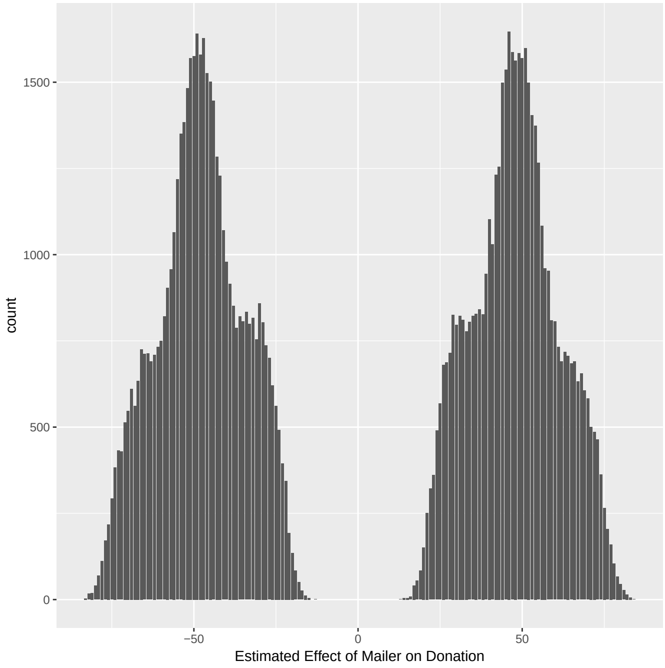
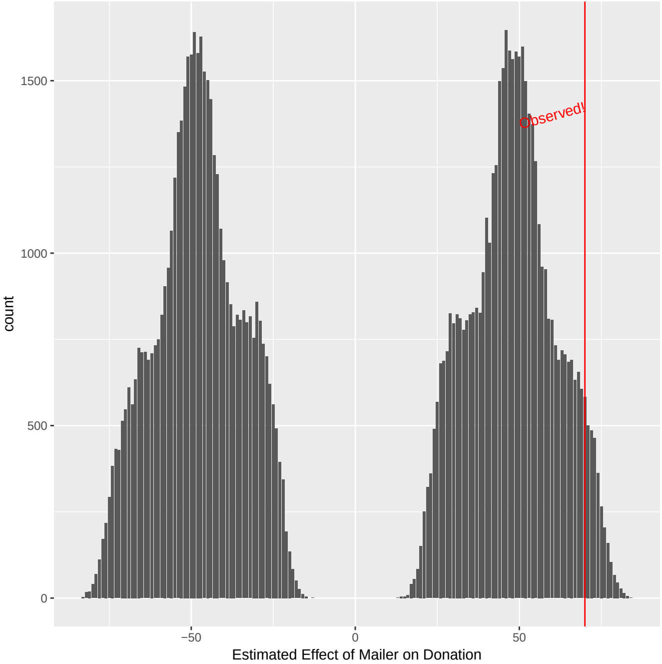
Randomisation Inference

- Gerber & Green donations example, p. 65
- Possible values $\tau_i \in (-\infty, \infty)$
- Y_1, Y_0, τ likely very skewed
- See `02-ri-resume-donate.R`


count
1500
1000
500
0
−50
0
50
Estimated Effect of Mailer on Donation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Recall that

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- See `02-ri-resume-donate.R`

Additional Resources

- https://methods.egap.org/guides/analysis-procedures/randomization-inference_en.html
- https://dimewiki.worldbank.org/Randomization_Inference

Next:
Covariates in Experiments
PS2 due

References I

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